

A Governance-First Architecture for Artificial Intelligence with Sealed Quantum Subroutines

Amos Jay Maley

Abstract

We present a governance-first artificial intelligence architecture in which quantum computation is permitted only as a sealed, non-authoritative subroutine for candidate generation. The architecture enforces strict separation between proposal generation and standing assignment, ensuring auditability, fail-closed operation, and immunity to optimizer authority escalation. We formalize the system layers, define the quantum interface contract, prove preservation of governance invariants, specify a mechanical conformance verification protocol, and derive a conditional necessity result. The result characterizes the maximal integration of quantum computation compatible with irreversible, audit-grade standing.

1 Introduction

Quantum computation has demonstrated the ability to generate candidate solutions to certain high-constraint problems more efficiently than known classical methods. However, naïve integration of quantum computation into artificial intelligence systems introduces unacceptable epistemic risks, including probabilistic authority leakage, non-auditable inference, and governance collapse through optimizer dominance.

This paper proposes a strict architectural remedy. We show that quantum computation may be safely integrated into AI systems only when it is subordinated to a governance layer that enforces admissibility, scope, and standing deterministically. The central claim is precise: *quantum computation may generate candidates, but it must never assign truth, authority, or standing.*

2 Governance-First Design Principles

The architecture is governed by the following non-negotiable principles:

1. Standing is binary and irreversible.¹
2. All authoritative assertions must be auditable.
3. No computational process may justify its own outputs.
4. Optimization is never governance.
5. Ambiguity is resolved by rejection.

¹Standing is not a belief measure, confidence score, probability, or policy preference. It is an authoritative commitment marker whose assignment is bivalent and irreversible per assertion.

2.1 Necessity of Authority Separation

The architecture asserts the following necessary condition:

Any system that permits probabilistic, feedback-driven, or optimizer-mediated standing assignment cannot provide fail-closed auditability under adversarial conditions.

Standing functions as an authoritative, irreversible commitment. If standing is graded, revisable, or conditioned on optimizer performance, authority becomes contingent on statistical behavior rather than admissibility, and rejection is no longer terminal. Such systems are categorically distinct from the class addressed here.

2.2 Binary Standing Requirement

Binary standing is a structural requirement. Standing is a commitment marker, not a confidence estimate. Any graded or probabilistic standing admits post hoc revision, invalidating fail-closed semantics by construction. Standing must therefore be bivalent, irreversible per assertion, and independent of optimizer performance.

3 System Architecture

The system is composed of three strictly ordered layers.

3.1 Layer 0: Governance Layer

The governance layer enforces admissibility prior to computation by:

- fixing task scope,
- assigning claim modality,
- declaring required witnesses,
- enforcing impossibility constraints.

The governance layer is deterministic and fail-closed.

3.2 Layer 1: Classical Reasoning Layer

This layer performs all authoritative reasoning:

- logical validation,
- consistency checking,
- proof construction,
- witness verification.

Standing is assigned exclusively here.

3.3 Layer 2: Quantum Subroutine Layer

Quantum computation is confined to candidate generation within pre-certified constraint envelopes. This layer:

- has no access to governance state,
- cannot assign standing,
- cannot modify constraints,
- retains no persistent memory.

4 Sealed Quantum Interface Contract

$$\text{Candidates} = QGenerate(\text{Constraints}, \text{Token})$$

- `Constraints` are governance-approved and immutable.
- `Token` is a non-authoritative capability handle.
- `Candidates` are provisional representations only.

4.1 Interface Invariants

1. No quantum output carries epistemic metadata.
2. No quantum output influences governance decisions.
3. No quantum state persists across invocations.
4. All outputs remain provisional until certified.

5 Safety Properties

5.1 Fail-Closed Semantics

Any ambiguity, scope violation, witness insufficiency, or impossibility detection results in immediate termination without output.

5.2 Authority Non-Escalation

Quantum subroutines cannot update rules, modify constraints, establish confidence, or justify conclusions.

5.3 Informational Non-Influence

Governance decisions are forbidden from depending on any statistical property of quantum execution, including success frequency, timing behavior, convergence rate, or repetition patterns. Informational side channels therefore cannot induce governance adaptation.

6 Quantitative Capability Assessment

Let T_g denote candidate generation time and T_v denote validation time, with $T_v \gg T_g$ in governed systems. Quantum subroutines reduce T_g by factors of 10^3 – 10^6 in suitable domains. End-to-end throughput improves modestly (~ 1 – $2\times$), while success probability in high-constraint landscapes increases substantially.

7 Engineering Complexity

The architecture presents moderate engineering difficulty:

- boundary enforcement: high discipline, known techniques,
- interface correctness: medium-high complexity,
- distributed orchestration: standard systems engineering,
- audit tooling: mature and well understood.

8 Conclusion

Quantum computation can be safely integrated into artificial intelligence systems only when denied epistemic authority. The architecture presented here characterizes the maximal quantum integration compatible with irreversible, audit-grade standing. The conditional necessity result derived in Appendix C applies specifically to systems requiring fail-closed auditability under adversarial conditions.

Appendix A: Formal Invariance Proofs

A.1 System State Decomposition

Let the global system state be:

$$S := (G, V, C, E)$$

where G is governance state, V validation state, C the candidate buffer, and E execution control state. Standing is a predicate definable only over (V, E) .

A.2 Governance Invariance

Theorem A.1. For all admissible execution traces, governance state G is invariant.

Proof. Transitions modifying G belong to Σ_G and occur only at ingress. Quantum execution induces transitions Σ_Q such that $(G, V, C, E) \rightarrow (G, V, C', E')$. Since $\Sigma_G \cap \Sigma_Q = \emptyset$, G is invariant. QED

A.3 Standing Non-Escalation

Theorem A.2. No element introduced by quantum execution can acquire standing.

Proof. Standing assignment requires a transition σ_S following modality assignment, validation, and witness verification. Quantum execution introduces candidates only via Σ_Q , which excludes σ_S . Thus standing cannot be acquired. QED

A.4 No Upward Influence

Theorem A.3. Quantum execution cannot influence governance evolution.

Proof. Governance transitions depend solely on G and task input. No transition references quantum statistics or timing. Hence no informational path exists from quantum execution to governance.

QED

A.5 Closure Preservation

Theorem A.4. No new admissible classes or repair strategies are introduced.

Proof. Admissibility is fixed in G . Since G is invariant and quantum execution operates only within certified envelopes, closure is preserved.

QED

Appendix B: Conformance Verification Protocol

B.1 Objective

To certify that implementations conform to architectural invariants prohibiting:

- non-classical standing assignment,
- post-ingress governance mutation,
- upward influence from quantum execution,
- violation of fail-closed semantics.

B.2 Model

Implementations are modeled as a transition system $\mathcal{S} = (Q, \Sigma, \delta, q_0)$ with state (G, V, C, E) .

B.3 Invariants

- $\mathbf{G}(G = G_0)$ after ingress,
- $\mathbf{G}(\forall c \in C : \neg \text{Standing}(c))$,
- $\mathbf{G}(\text{Ambiguity} \Rightarrow \text{Halt})$.

B.4 Conformance Conditions

An implementation conforms iff:

1. all G mutations occur only at ingress,
2. standing is assigned only via Σ_S ,
3. no transition references quantum statistics,
4. all ambiguity states terminate.

Functional correctness, optimization quality, and performance are explicitly out of scope.

Appendix C: Conditional Necessity Result

Theorem C.1 (Audit-Grade Necessity). Any AI system that assigns standing probabilistically, revisably, or as a function of optimizer behavior cannot satisfy fail-closed auditability under adversarial conditions.

Proof. Fail-closed auditability requires terminal rejection and irreversible authority assignment. Probabilistic or revisable standing violates irreversibility. Optimizer-conditioned standing makes authority contingent on non-admissible statistics. In all cases, auditability fails. QED